

SOFTWARE TESTING & VALIDATION · FINAL PROJECT

EasyService: Test Summary Report & Foundations Reflection — Parts H & I (v1.0)

Institution: Addis Ababa University — School of Information Technology and Engineering

Course: Software Testing and Validation

Document: Test Summary Report & Foundations Reflection (Parts H & I)

Repository: <https://github.com/Abyman2/Easyservice>

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1. Executive Summary & Release Recommendation

This Test Summary Report evaluates the software quality, coverage, defect metrics, and release readiness of **EasyService v1.0**.

Based on complete execution of the test plan:

- 1. **100% of planned in-scope features** pass unit and integration tests.
- 2. **JaCoCo Branch Coverage** reached **86.4%**, exceeding the 80.0% threshold.
- 3. **Defect Removal Efficiency (DRE)** is **100%** with zero open Critical or High severity defects.
- 4. **CI/CD Automation Pipelines** (GitHub Actions & Jenkins) run successfully.

FINAL RECOMMENDATION: ACCEPT & RELEASE READY FOR DEPLOYMENT

2. Testing Pyramid & Level Summary

- **Unit Testing (JUnit 5 + Mockito):** 35 unit tests validating business rules BR-01 to BR-20, equivalence partitions, boundary values, and decision tables.
 - **Integration Testing (Spring Boot Test + MockMvc):** 6 integration scenarios verifying REST controller execution and in-memory persistence workflows.
 - **System Testing (Selenium WebDriver):** Page Object Model journeys covering registration, verification, search, booking, and cancellation.
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3. Foundations Reflection & Concept Synthesis

3.1 Verification vs. Validation

- **Verification:** "Are we building the product right?" Ensured through static code analysis, JaCoCo coverage enforcement, and unit tests adhering to BR-01..BR-20.
- **Validation:** "Are we building the right product?" Ensured through Selenium end-to-end user journeys validating real customer booking and provider listing workflows.

3.2 Error, Fault, Failure Spectrum

- **Error:** Developer misunderstanding of overbooking boundary conditions.
- **Fault:** Bug in `BookingTransactionService.java` using `>=` instead of `>`.
- **Failure:** System accepting booking quantity 101 when capacity was 100 during automated test run.

3.3 Test Doubles Selection Rationale

- **Fake:** `FakePaymentService` and `FakeIdentityVerificationService` provided full in-memory working logic with deterministic outcome toggling without real financial/identity exposure.
- **Mock/Spy:** `FakeNotificationService` tracked invocation counts to verify notification dispatches.
- **Controlled Randomness:** `FakeRandomNumberGenerator` eliminated non-deterministic randomness in the Wheel tool to enable reproducible test assertions.